

AWS Training Guide



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Identity:

What is AWS IAM?

Amazon IAM (Identity and Access Management) is a service provided by Amazon Web Services (AWS) that enables users to manage access to AWS services and resources securely. It helps control who (users) can access what (AWS resources) and what actions they can perform.



Key features and functionalities of Amazon IAM include:

- 1. User Management:** IAM allows users to create and manage AWS users and groups, defining their permissions to access specific AWS services and resources.
- 2. Access Control:** Users can define fine-grained access policies using IAM policies, which specify permissions for users, groups, or roles to perform specific actions on AWS resources.
- 3. Multi-Factor Authentication (MFA):** IAM supports MFA, providing an extra layer of security by requiring users to present two or more authentication factors (such as a password and a temporary code from a mobile device) to access AWS resources.
- 4. Roles and Temporary Credentials:** IAM roles allow users or services to assume temporary permissions to access AWS resources securely without the need to share long-term credentials.
- 5. Integration with AWS Services:** IAM integrates seamlessly with various AWS services, allowing users to control access to services like EC2 instances, S3 buckets, RDS databases, and more.
- 6. Auditing and Monitoring:** IAM provides detailed logs through AWS CloudTrail, allowing users to monitor and audit actions performed by users, including changes to policies, roles, or resource access.

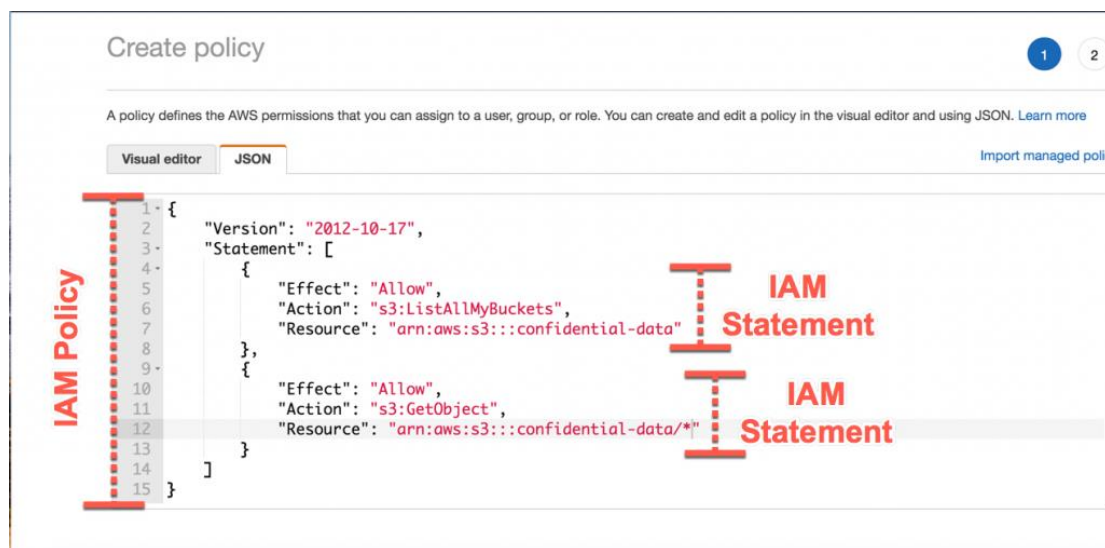
IAM is crucial for implementing security best practices within an AWS environment. By managing identities and their permissions effectively, it helps organizations enforce the principle of least

privilege, ensuring that users have only the necessary access required to perform their tasks while maintaining overall security and compliance.

Amazon Identity and Access Management allows:

- ❖ users to analyse access and provide MFA (Multi-factor authentication) to protect the AWS environment.
- ❖ managing IAM users, IAM roles, and federated users.

1. IAM Policies: These are sets of permissions that define what actions are allowed or denied on AWS resources. Policies are written in JSON format and can be attached to IAM users, groups, or roles. They specify the permissions that individuals or entities have within an AWS environment.



2. IAM Users: These are entities within AWS accounts that interact with AWS resources. Each user has specific security credentials (such as a username and password or access keys) and associated permissions. Users can be granted permissions directly or through membership in IAM groups.

3. IAM Groups: These are collections of IAM users. Groups make it easier to manage permissions for multiple users, as permissions can be assigned to the group as a whole rather than to individual users. Users can belong to multiple groups, and their permissions are the combination of all the policies attached to the groups they are part of.

4. IAM Roles: Roles are a way to delegate access to users, applications, or services that may not normally have access to certain AWS resources. They are used to define a set of permissions that a user or service can assume temporarily. Roles are commonly used in scenarios like granting permissions to an AWS service or enabling cross-account access.

IAM policies define what actions are allowed or denied, users are the entities interacting with AWS, groups organize users for easier permission management, and roles facilitate controlled access for entities or services that need temporary access to resources.

By effectively using IAM, AWS account owners can ensure that only the right people or services have access to the right resources at the right time.

Storage:

What is Amazon Simple Storage Service?

Amazon Simple Storage Service (S3) is a scalable, durable, and secure object storage service provided by Amazon Web Services (AWS). It's designed to store and retrieve any amount of data from anywhere on the web, making it highly versatile and widely used for various storage needs.

Storage



Key features of Amazon S3 include:

- 1. Scalability:** S3 allows users to store virtually unlimited amounts of data, and it automatically scales to accommodate storage needs without requiring manual intervention.
- 2. Durability and Availability:** S3 is designed for 99.999999999% (11 nines) of durability and offers high availability, ensuring that stored data remains accessible and protected from potential failures.
- 3. Security and Access Control:** Users can control access to their data using Access Control Lists (ACLs), bucket policies, and Identity and Access Management (IAM) policies. Encryption options, including server-side encryption, enhance data security.
- 4. Data Management:** S3 provides features for managing data, such as versioning, lifecycle policies (for automating data migration or deletion based on predefined rules), and cross-region replication for redundancy.

5. Performance and Speed: It offers high-speed data transfer and low-latency access to stored objects, making it suitable for a wide range of use cases, including static website hosting, content delivery, backup and archiving, data lakes, and more.

6. Integration and Compatibility: S3 integrates seamlessly with various AWS services, third-party tools, and applications, making it easy to use as a storage backend for a diverse set of workflows and applications.

Amazon S3's simple and intuitive RESTful APIs make it accessible for developers to integrate storage capabilities into their applications, while its reliability, scalability, and security make it a fundamental building block for many cloud-based architectures and data storage solutions.

Amazon S3 provides the following storage classes used to maintain the integrity of the objects:

- ☐ S3 Standard - offers frequent data access.
- ☐ S3 Intelligent-Tiering - automatically transfer data to other cost-effective access tiers.
- ☐ S3 Standard-IA - offers immediate and infrequent data access.
- ☐ S3 One Zone-IA - infrequent data access.
- ☐ S3 Glacier - long-term archive data, cheap data retrieval.
- ☐ S3 Glacier Deep Archive - used for long-term retention.

Amazon S3 offers to choose from the following ways to replicate objects:

- **Cross-Region Replication** - used to replicate objects in different AWS Regions.
- **Same Region Replication** - used to replicate objects in the same AWS Region.

Amazon S3 Glacier is a web service with vaults that offer long-term data archiving and data backup.

S3 Glacier provides the following data retrieval options:

Expedited retrievals:

- It retrieves data in 1-5 minutes.

Standard retrievals:

- It retrieves data between 3-5 hours.

Bulk retrievals:

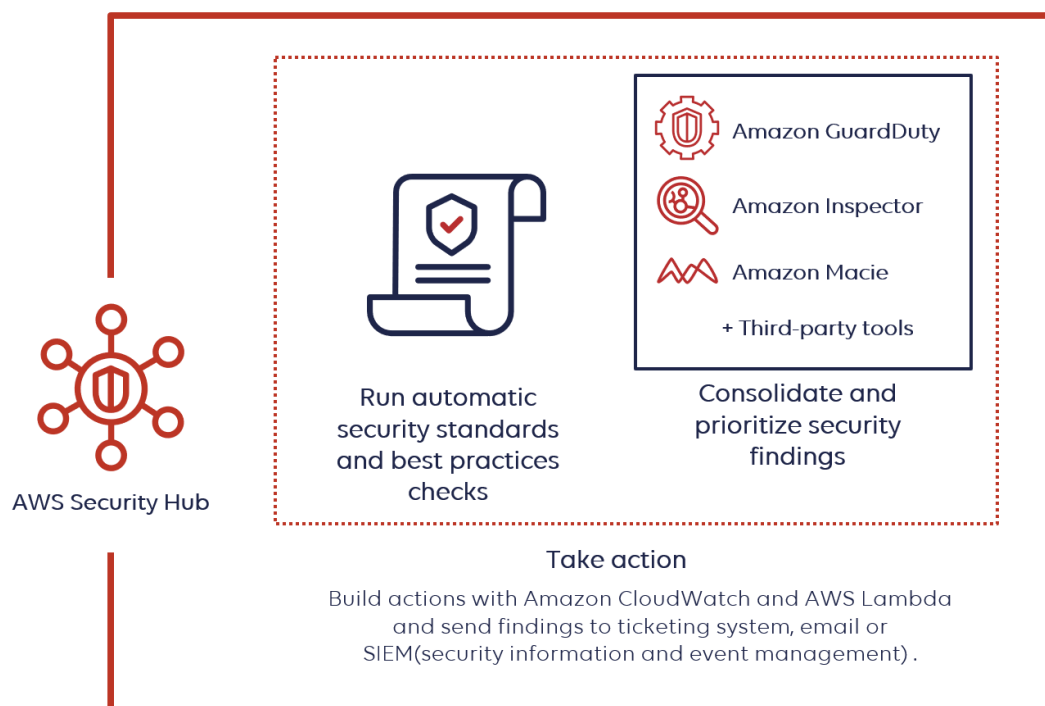
- It retrieves data between 5-12 hours.

Compliance:

What is Security Hub?

AWS Security Hub is a comprehensive security service provided by Amazon Web Services (AWS) that offers a centralized view of security alerts and compliance status across an AWS environment. It aggregates, organizes, and prioritizes security findings from various AWS services and supported third-party security tools, providing a comprehensive overview of the security posture.

AWS Security Hub helps the Payment Card Industry Data Security Standard (PCI DSS) and the Centre for Internet Security (CIS) AWS Foundations Benchmark with a set of security configuration best practices for AWS.



Key features and functionalities of AWS Security Hub include:

1. Aggregated Security Insights: Security Hub collects and consolidates security findings and alerts from multiple AWS services such as Amazon Guard Duty, Amazon Inspector, AWS Firewall Manager, and more. It also integrates with select third-party security tools.

2. Security Standards and Best Practices: It helps users adhere to security best practices and compliance standards by providing predefined security standards (such as AWS Foundational Security Best Practices, CIS AWS Foundations Benchmark) and continuously monitoring compliance status.

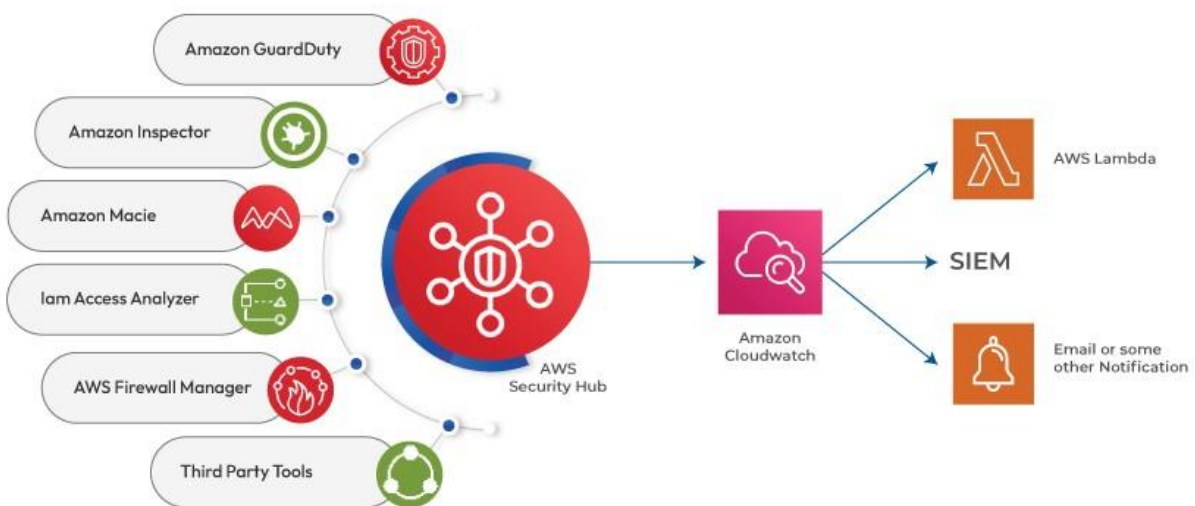
3. Prioritized Findings: Security Hub prioritizes security findings based on severity levels, providing insights into the most critical issues that require immediate attention.

4. Workflow Integration: It integrates with AWS services like Amazon CloudWatch Events, AWS Lambda, and AWS Systems Manager to automate response actions, notifications, and remediation workflows based on security findings.

5. Custom Insights and Aggregation: Users can create custom insights and aggregation based on specific criteria, allowing them to focus on relevant security issues and trends within their environment.

6. Visibility and Remediation: It offers a consolidated dashboard and graphical representations to visualize security trends, making it easier for security teams to identify, investigate, and remediate security issues.

AWS Security Hub simplifies security management by providing a single pane of glass for monitoring the overall security posture of an AWS environment. It helps organizations improve their security and compliance by identifying and addressing potential vulnerabilities and security misconfigurations across their cloud infrastructure.



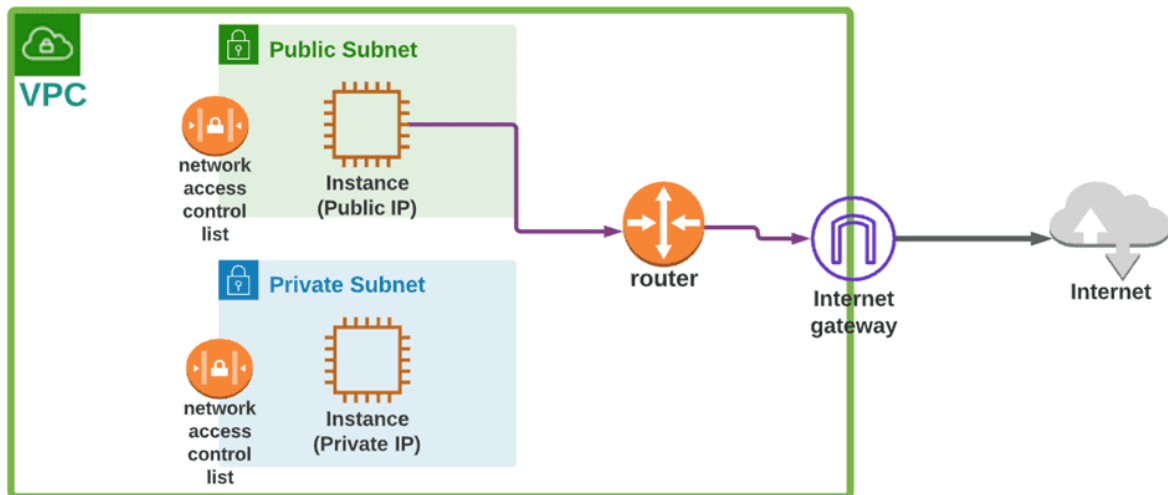
- AWS Security Hub provides an option to aggregate, organize, and prioritize the security alerts or findings from multiple AWS services.
- It automatically checks the compliance status using CIS AWS Foundations Benchmark.
- It uses integrated dashboards to show the current security and compliance status.
- It collects findings or alerts from multiple AWS accounts. Then it analyses security trends and identifies the highest priority security issues.
- The security alerts or findings can be investigated using Amazon Detective or Amazon CloudWatch Event rules.

It collects data from AWS services across accounts and reduces the need for time-consuming data conversion efforts. Charges are applied only for the current Region, not for all Regions in which Security Hub is enabled.

Networking:

What is AWS VPC?

AWS VPC (Virtual Private Cloud) is a service that allows users to create and manage a logically isolated section of the Amazon Web Services (AWS) cloud. It enables users to launch AWS resources in a virtual network that closely resembles a traditional network setup, offering control over the network environment, including IP address ranges, subnets, routing tables, and network gateways.



Key components and features of AWS VPC include

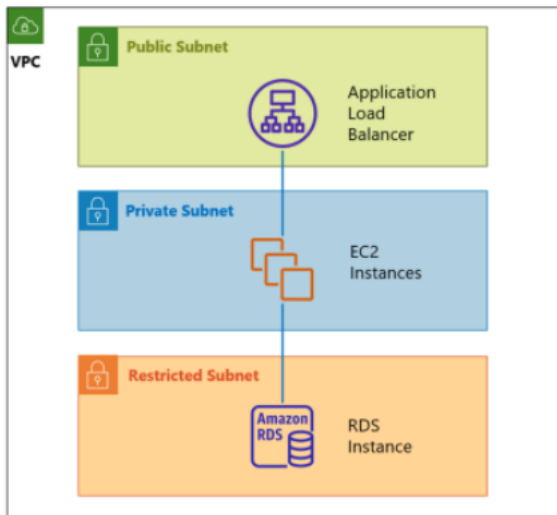
- 1. Isolation and Control:** VPC provides isolation for resources, allowing users to define their virtual network environment with complete control over IP addressing, subnets, routing, and security settings.
- 2. Subnet:** Users can divide the VPC into subnets to organize resources, and each subnet can be associated with an availability zone, providing high availability and fault tolerance.
- 3. Routing:** Users can define routing tables to control traffic between subnets, direct traffic to specific destinations, or connect the VPC to other networks.
- 4. Internet Gateways and NAT Gateways:** Internet Gateways enable communication between instances in the VPC and the internet, while NAT Gateways allow instances in private subnets to access the internet while keeping them private.
- 5. Security:** VPC allows users to implement security measures using security groups and network access control lists (ACLs) to control inbound and outbound traffic.
- 6. VPN and Direct Connect:** VPC enables secure connections between the VPC and the corporate data centre or other networks using VPN (Virtual Private Network) or AWS Direct Connect.

AWS VPC gives users the flexibility to build a virtual network environment that meets their specific requirements, whether it's hosting a simple web application or creating a complex, multi-tiered architecture. It's a foundational service that forms the basis for deploying various AWS resources in a controlled and secure network setup.

Private subnet - A subnet that does not have internet access is termed a private subnet.

Public subnet - A subnet that has internet access is termed a public subnet.

VPN only subnet (Restricted subnet) - A subnet that does not have internet access but has access to the virtual private gateway for a VPN connection is termed a VPN-only subnet.



- ❑ It includes many components such as Internet gateways, VPN tools, CIDR, Subnets, Route tables, VPC endpoint, NAT instances, Bastion servers, Peering Connection, and others.
- ❑ It spans across multiple Availability Zones (AZs) within a region.
- ❑ The first four IP and last one IP addresses are reserved per subnet.
- ❑ It creates a public subnet for web servers that uses internet access and a private subnet for backend systems, such as databases or application servers.
- ❑ It can monitor resources using Amazon CloudWatch and Auto Scaling Groups.
- ❖ Every EC2 instance is launched within a default VPC with equal security and control like normal Amazon VPC. Default VPC has no private subnet.
- ❖ It uses Security Groups and NACL (Network Access Control Lists) for multi-layer security.
- ❖ Security Groups (stateful) provide instance-level security, whereas NACLs (stateless) provide subnet-level security.
- ❖ VPC sharing is a component that allows subnets to share with other AWS accounts within the same AWS Organization.

What is Amazon CloudFront?

Amazon CloudFront is a content delivery network (CDN) service provided by Amazon Web Services (AWS). It's designed to deliver data, videos, applications, and APIs to users globally with low latency, high transfer speeds, and reduced load times by caching content at edge locations around the world.



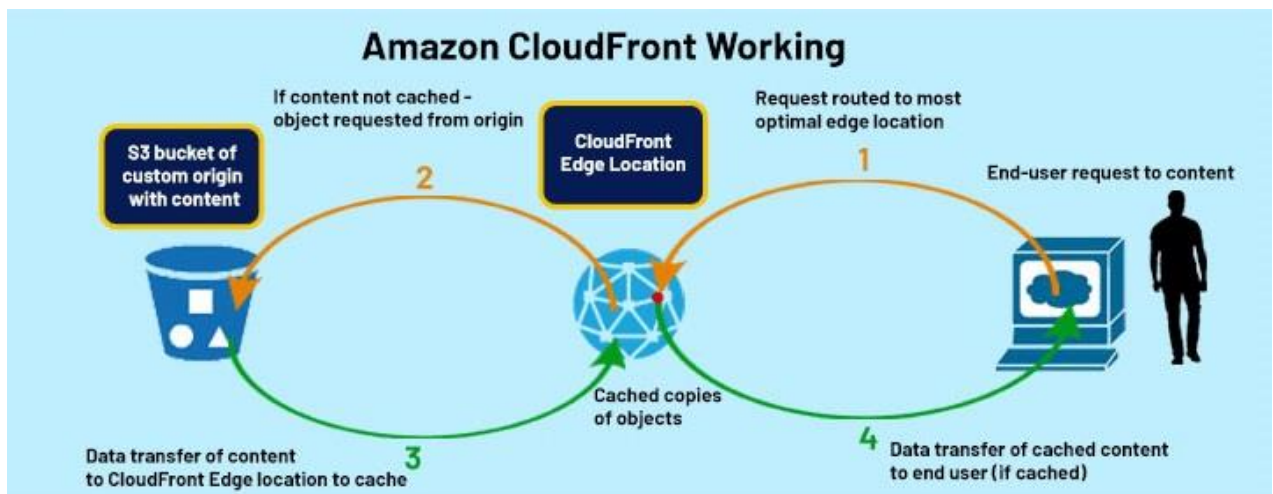
Key features of Amazon CloudFront include:

- 1. Content Distribution:** CloudFront caches content at edge locations (data centres situated in various geographical locations), allowing users to access data from a location closer to them, thereby reducing latency.
- 2. Global Reach:** It has a widespread network of edge locations across multiple continents, enabling fast and reliable content delivery to end-users regardless of their location.
- 3. Security:** CloudFront integrates with other AWS services like AWS Shield for DDoS mitigation and AWS Web Application Firewall (WAF) to protect against various security threats.
- 4. Customization:** Users can customize settings for cache control, origin configuration, SSL/TLS protocols, and more to suit their specific requirements.
- 5. Real-Time Monitoring:** CloudFront provides real-time monitoring and metrics through AWS CloudWatch, allowing users to track the performance and behaviour of their content delivery.
- 6. Integration with AWS Services:** It seamlessly integrates with other AWS services like Amazon S3 (Simple Storage Service), EC2 (Elastic Compute Cloud), and Elastic Load Balancing, allowing easy setup and management.

CloudFront is commonly used to deliver static and dynamic web content, streaming media, APIs, and even entire websites. By caching content closer to users at edge locations, it enhances the overall user experience by reducing latency and improving the responsiveness of web applications and services.

It makes use of Edge locations (worldwide network of data centres) to deliver the content faster. [?] Without edge locations, it retrieves data from an origin such as an Amazon S3 bucket, a Media Package channel, or an HTTP server.

Amazon EC2
Amazon S3
Elastic Load Balancing
Amazon Route 53
CloudFront is integrated with AWS Services such as:
AWS Essential Media Services



CloudFront provides some security features such as:

- ❖ Field-level encryption with HTTPS - Data remains encrypted throughout starting from the upload of sensitive data.
- ❖ AWS Shield Standard - Against DDoS attacks.
- ❖ AWS Shield Standard + AWS WAF + Amazon Route 53 – Against more complex attacks than DDoS.

Amazon CloudFront Access Controls:

Signed URLs:

- Use this to restrict access to individual files.

Signed Cookies:

- Use this to provide access to multiple restricted files.
- Use this if the user does not want to change current URLs.

Geo Restriction:

- Use this to restrict access to the data based on the geographic location of the website viewers.

Origin Access Identity (OAI):

- Outside access is restricted using signed URLs and signed cookies, but what if someone tries to access objects using Amazon S3 URL, bypassing CloudFront signed URL and signed cookies. To restrict that, OAI is used.
- Use OAI as a special CloudFront user and associate it with your CloudFront distribution to secure Amazon S3 content.

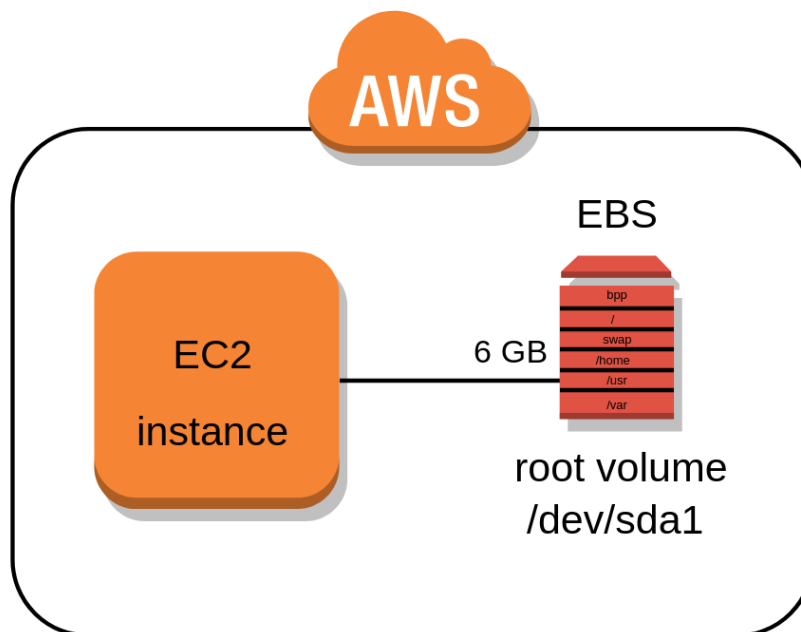
Compute:

What is AWS EC2?

Amazon Elastic Compute Cloud (Amazon EC2) is a service that provides secure and scalable compute capacity in the AWS cloud. It falls under the category of Infrastructure as a Service (IAAS).

AWS EC2 (Elastic Compute Cloud) is a web service provided by Amazon Web Services that allows users to rent virtual machines (EC2 instances) on which they can run their applications. It's one of the foundational services offered by AWS and is designed to provide scalable compute capacity in the cloud.

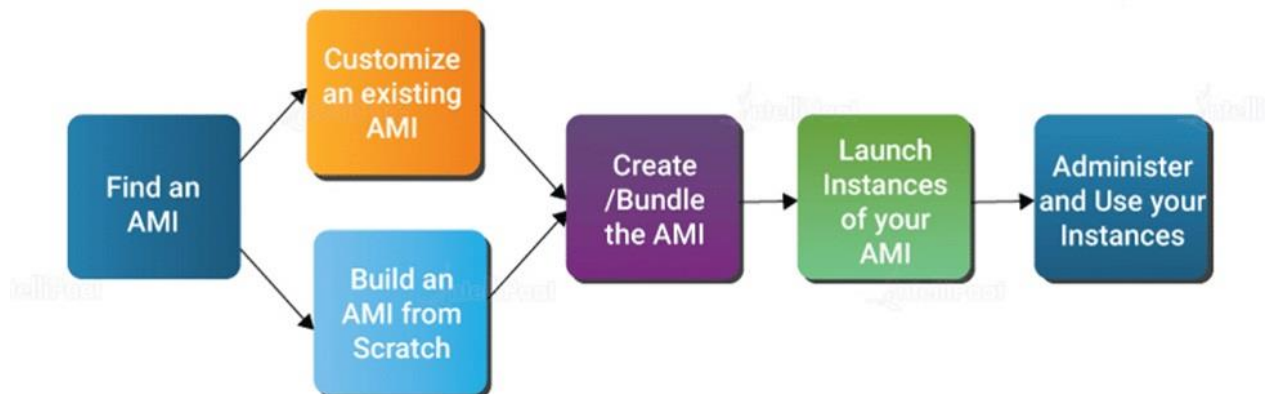
EC2 instances are virtual servers that can be configured with varying computational power, memory, storage, and networking capacity based on the user's needs E.g., t2.micro, t4g.nano, m4.large, r5a.large, etc.. Users can choose from a wide range of pre-configured instance types or create custom configurations to suit their specific requirements.



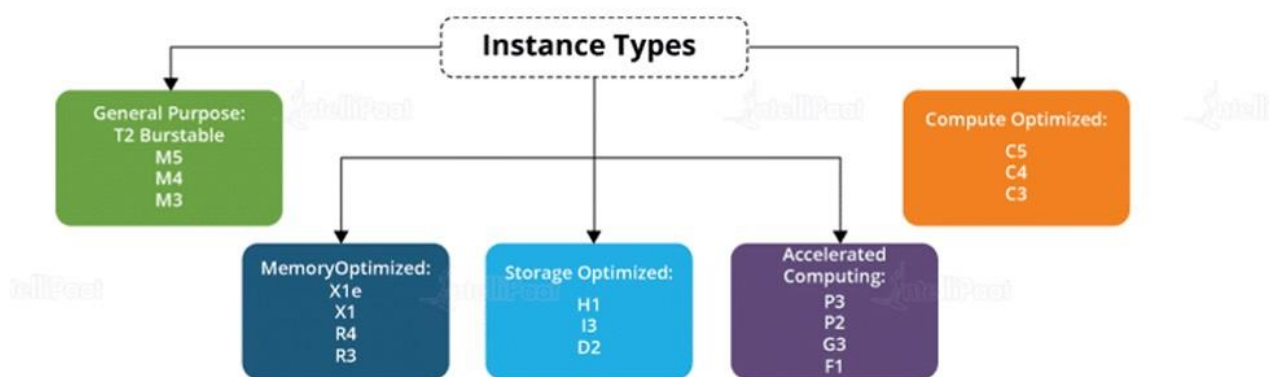
Key features of AWS EC2 include:

- 1. Scalability:** Users can easily scale the number of instances up or down based on demand, allowing flexibility in handling varying workloads.
- 2. Various Instance Types:** AWS offers a variety of instance types optimized for different use cases, such as compute-optimized, memory-optimized, storage-optimized, etc.
- 3. Customization:** Users have control over the operating system, networking, security settings, and storage options for their EC2 instances.
- 4. Pricing Models:** EC2 offers different pricing models, including on-demand (pay-as-you-go), reserved instances (for a discounted rate with a commitment), and spot instances (allowing users to bid for unused capacity).

5. Integration and Services: It integrates seamlessly with other AWS services like Elastic Block Store (EBS) for storage, Elastic Load Balancing (ELB) for load balancing, and Amazon VPC for networking, among others.



EC2 is widely used by businesses and developers to deploy applications, host websites, run batch processing jobs, perform data analysis, and more, providing the infrastructure backbone for various cloud-based computing needs.



Types of instances based on the pricing models:

1. **On-Demand Instances**
 - ✓ Useful for short-term needs, unpredictable workloads.
 - ✓ No advance payment, no prior commitment.
2. **Spot Instances**
 - ✓ No advance payment, no prior commitment.
 - ✓ Useful for cost-sensitive compute workloads.
3. **Reserved Instances**
 - ✓ Useful for long-running workloads and predictable usage.
 - ✓ Offer to choose from No upfront, Partial upfront, or All upfront.
4. **Dedicated Instances**
 - ✓ Instances run on hardware dedicated to a single user.
 - ✓ Other customers can not share the hardware.
5. **Dedicated Hosts**
 - ✓ A whole physical server with an EC2 instance allocates to an organization

EC2 INSTANCE PRICING OPTIONS			
	ON-DEMAND	RESERVED	SPOT
BILLING PERIOD	Hourly	Contract, paid upfront, partial upfront or no upfront	Hourly
PRICE	AWS specified hourly rate	Up to 75% off hourly rate. Can also purchase from user marketplace	Up to 90% off hourly rate. Bidding process
TERM	No commitment, used as needed	1-year or 3-year contracts if purchased from AWS. Varies in marketplace	Instance stops when bid exceeds customer's maximum bid
RECOMMENDED FOR	Unpredictable workloads, applications being tested in EC2	Applications with steady usage or need reserved capacity	Applications with flexible start/end times. Low-cost projects. Workloads that urgently need extra capacity

- It provides pre-configured templates that package the operating system and other software for the instances. This template is called Amazon Machine Images (AMIs).
- It helps to login into the instances using key-pairs, in which AWS manages the public key, and the user operates the private key.
- It enables users to write scripts under the option 'User data,' used at the instances' launch.
- It also provides firewall-like security by specifying IP ranges, type, protocols (TCP), port range (22, 25, 443) using security groups.
- It provides temporary storage volumes known as instance store volumes, which are deleted if the instance gets stopped, hibernated, or terminated. It also offers non-temporary or persistent volumes known as Amazon EBS volumes.
- It offers to choose from three IP addresses, which are Public IP address (Changes when the instance is stopped or refreshed), Private IP address (retained even if the model is stopped), Elastic IP address (static public IP address).

EC2 Auto Scaling:

Amazon EC2 Auto Scaling is a region-specific service used to maintain application availability and enables users to automatically add or remove EC2 instances according to the compute workloads.

- ❖ The Auto Scaling group is a collection of the minimum number of EC2 used for high availability.
- ❖ It enables users to use Amazon EC2 Auto Scaling features such as fault tolerance, health check, scaling policies, and cost management.
- ❖ The scaling of the Auto Scaling group depends on the size of the desired capacity. It is not necessary to keep Desired Capacity and MaxSize equal.
E.g., Desired Capacity: '2' - There will be total 2 EC2 instances MinSize: '1' MaxSize: '2'
- ❖ EC2 Auto Scaling supports automatic Horizontal Scaling (increases or decreases the number of EC2 instances) rather than Vertical Scaling (increases or decreases EC2 instances like large, small, medium).

Auto Scaling Groups:

- Manual Scaling
- Update the desired.
- capacity of the Auto
- Scaling Group
- manually.

Scheduled Scaling:

This scaling policy adds or removes instances based on predictable traffic patterns of the application.

Dynamic Scaling:

- **Target tracking scaling policy:** This scaling policy adds or removes instances to keep the scaling metric close to the specified target value.
- **Simple Scaling Policy:** This scaling policy adds or removes instances when the scaling metric value exceeds the threshold value.
- **Step Scaling Policy:** This scaling policy adds or removes instances based on step adjustments (lower bound and upper bound of the metric value).

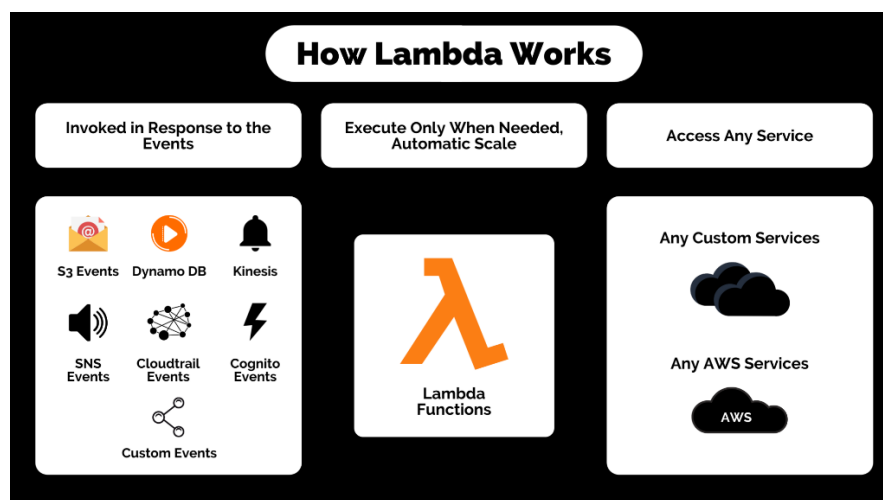
Cooldown Period:

The Cooldown period is the time during which an Auto Scaling group doesn't launch or terminate any instances before the previous scaling activity completes.

What is AWS Lambda?

AWS Lambda is a serverless compute service provided by Amazon Web Services (AWS) that allows you to run code without provisioning or managing servers. It operates on a pay-as-you-go model, charging you only for the compute time consumed by your code.

Lambda lets you upload your code (written in languages like Node.js, Python, Java, etc.) and then automatically manages the infrastructure needed to execute that code in response to specific events. These events can originate from various AWS services, HTTP requests via Amazon API Gateway, or even custom triggers.



One of Lambda's significant advantages is its scalability and cost-effectiveness. It automatically scales based on the incoming requests, running your code in response to each trigger, and you don't have to worry about the underlying infrastructure. This makes it particularly useful for building applications with highly variable workloads or event-driven architectures.

Lambda Functions supports the following languages:

- Java Power
- shell
- Go
- Python
- Node.js
- Ruby
- C#

Pricing Details:

Charges are applied based on the number of requests for the functions and the time taken to execute the code.

Memory:

The memory allocated to AWS Lambda for computing is 128MB (minimum) to 3008MB (maximum). Additional memory can be requested in an increment of 64MB between 128MB - 3008MB.

Execution Time:

The default execution time for AWS Lambda is 3 seconds, and the maximum is 15 minutes (900 seconds).